

2520090016

P.Rahul

WEEK-4

BOOKFLOW INTELLIGENT CATALOG & API BLUEPRINT

The screenshot displays the MySQL Workbench interface. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The left sidebar shows the 'SCHEMAS' tab with a list of databases: 2520090016_rahul, bankdb, joins_set_operations, klu, rahul_2520090016, sys, and workflow_db. The main editor window, titled 'SQL File 6*', contains the following SQL script:

```
1 • DROP DATABASE IF EXISTS bookflow_db;  
2 • CREATE DATABASE bookflow_db;  
3 • USE bookflow_db;  
4
```

The bottom section of the interface shows the 'Output' tab with the 'Action Output' view. It displays a table of execution results:

#	Time	Action	Message
✓ 1	14:32:39	DROP DATABASE IF EXISTS bookflow_db	3 row(s) affected
✓ 2	14:32:57	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 3	14:33:13	USE bookflow_db	0 row(s) affected

SQL File 6

```
1 • DROP DATABASE IF EXISTS bookflow_db;
2 • CREATE DATABASE bookflow_db;
3 • USE bookflow_db;
4 • SELECT DATABASE();
```

Result Grid

DATABASE()
bookflow_db

Schema: 2520090016_rahul

Result 1

Output

Action Output

#	Time	Action	Message
✓ 1	14:32:39	DROP DATABASE IF EXISTS bookflow_db	3 row(s) affected
✓ 2	14:32:57	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 3	14:33:13	USE bookflow_db	0 row(s) affected
✓ 4	14:40:53	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned

PART 2 — Create the Books table

MySQL Workbench interface showing the creation of a database and a table.

Navigator:

- 2520090016_rahul
 - bankdb
 - joins_set_operations
 - klu
 - rahul_2520090016
 - sys
 - workflow_db

SQL File 6:

```
1 DROP DATABASE IF EXISTS bookflow_db;
2 CREATE DATABASE bookflow_db;
3 USE bookflow_db;
4 SELECT DATABASE();
5 CREATE TABLE Books
6 (
7     book_id INT AUTO_INCREMENT PRIMARY KEY,
8     title VARCHAR(200) NOT NULL,
9     author VARCHAR(150) NOT NULL,
10    genre VARCHAR(100),
11    description TEXT,
12    published_year INT
13 )
```

Administration Schemas

Information

Schema:
2520090016_rahul

Output

Action Output

#	Time	Action	Message
✓ 1	14:32:39	DROP DATABASE IF EXISTS bookflow_db	3 row(s) affected
✓ 2	14:32:57	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 3	14:33:13	USE bookflow_db	0 row(s) affected
✓ 4	14:40:53	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
✓ 5	14:41:56	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NU...	0 row(s) affected

Object Info **Session**

MySQL Workbench

Rahul_2520090016

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- 2520090016_rahul
- bankdb
- joins_set_operations
- klu
- rahul_2520090016
- sys
- workflow_db

SQL File 6

```

8      title VARCHAR(200) NOT NULL,
9      author VARCHAR(150) NOT NULL,
10     genre VARCHAR(100),
11     description TEXT,
12     published_year INT
13 );
14 • DESCRIBE Books;

```

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	<u>NULL</u>	auto_increment
title	varchar(200)	NO		<u>NULL</u>	
author	varchar(150)	NO		<u>NULL</u>	
genre	varchar(100)	YES		<u>NULL</u>	
description	text	YES		<u>NULL</u>	
published_year	int	YES		<u>NULL</u>	

Administration Schemas

Information

Schema: 2520090016_rahul

Result 2

Output

Action Output

#	Time	Action	Message
✓ 1	14:32:39	DROP DATABASE IF EXISTS bookflow_db	3 row(s) affected
✓ 2	14:32:57	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 3	14:33:13	USE bookflow_db	0 row(s) affected
✓ 4	14:40:53	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
✓ 5	14:41:56	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NU...	0 row(s) affected
✓ 6	14:42:59	DESCRIBE Books	6 row(s) returned

Object Info Session

PART 3 — Insert the books

MySQL Workbench

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 6

Limit to 1000 rows

SCHEMAS

Filter objects

- 2520090016_rahul
 - bankdb
 - joins_set_operations
 - klu
 - rahul_2520090016
 - sys
 - workflow_db

Administration Schemas

Information

Schema: 2520090016_rahul

```
19 genre,  
20 description,  
21 published_year  
22 )  
23 VALUES  
24 (  
25 'The Martian',  
26 'Andy Weir',  
27 'Science Fiction',  
28 'An astronaut struggles to survive alone on Mars.',  
29 2011  
30 ),  
31 (  
32 'Dune',  
33 'Frank Herbert',  
34 'Science Fiction',  
35 'A story involving planets, politics, survival and space.',  
36 1965  
37 ),  
38 (  
39 'The Hitchhikers Guide to the Galaxy',  
40 'Douglas Adams',  
41 'Science Fiction',
```

Output

Action Output

#	Time	Action	Message
2	14:32:57	CREATE DATABASE bookflow_db	1 row(s) affected
3	14:33:13	USE bookflow_db	0 row(s) affected
4	14:40:53	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
5	14:41:56	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
6	14:42:59	DESCRIBE Books	6 row(s) returned
7	14:44:26	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

Object Info Session

MySQL Workbench

Rahul_2520090016

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 6*

Limit to 1000 rows

88 'Twenty Thousand Leagues Under the Seas',
 89 'Jules Verne',
 90 'Adventure',
 91 'A fantastic underwater adventure aboard a mysterious submarine.',
 92 1870
 93);
 94 SELECT * FROM Books;

Result Grid

book_id	title	author	genre	description	published_year
1	The Martian	Andy Weir	Science Fiction	An astronaut struggles to survive alone on Mars.	2011
2	Dune	Frank Herbert	Science Fiction	A story involving planets, politics, survival and ...	1965
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	A humorous adventure across space and alien ...	1979
4	Pride and Prejudice	Jane Austen	Romance	A classic story about relationships and social ex...	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	A story of ambition, wealth, love and the Ameri...	1925
6	Foundation	Isaac Asimov	Science Fiction	A futuristic story about mathematics, civilization...	1951
7	The Hobbit	J.R.R. Tolkien	Fantasy	A fantasy adventure involving Bilbo Baggins an...	1937
8	Enders Game	Orson Scott Card	Science Fiction	A young student is trained for an interstellar co...	1985
9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	A young wizard begins his magical education.	1997
10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	A fantastic underwater adventure aboard a my...	1870

Schema: 2520090016_rahul

Books 3

Output

#	Time	Action	Message
3	14:33:13	USE bookflow_db	0 row(s) affected
4	14:40:53	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
5	14:41:56	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
6	14:42:59	DESCRIBE Books	6 row(s) returned
7	14:44:26	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
8	14:44:57	SELECT * FROM Books LIMIT 0, 1000	10 row(s) returned

PART 4 — Add the embedding column

MySQL Workbench

Rahul_2520090016

File Edit View Query Database Server Tools Scripting Help

Navigator: SQL File 6~

Limit to 1000 rows

SCHEMAS

Filter objects

- 2520090016_rahul
 - bankdb
 - joins_set_operations
 - klu
 - rahul_2520090016
 - sys
 - workflow_db

Administration Schemas

Information

Schema: 2520090016_rahul

```

75 'Orson Scott Card',
76 'Science Fiction',
77 'A young student is trained for an interstellar conflict.',
78 1985
79 ),
80 (
81 'Harry Potter and the Sorcerers Stone',
82 'J.K. Rowling',
83 'Fantasy',
84 'A young wizard begins his magical education.',
85 1997
86 ),
87 (
88 'Twenty Thousand Leagues Under the Seas',
89 'Jules Verne',
90 'Adventure',
91 'A fantastic underwater adventure aboard a mysterious submarine.',
92 1870
93 );
94 SELECT * FROM Books;
95 ALTER TABLE Books
96 ADD COLUMN embedding JSON;

```

Output

Action Output

#	Time	Action	Message
4	14:40:53	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
5	14:41:56	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
6	14:42:59	DESCRIBE Books	6 row(s) returned
7	14:44:26	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian...	10 row(s) affected Records: 10 Duplicates: 0 Warning
8	14:44:57	SELECT * FROM Books LIMIT 0, 1000	10 row(s) returned
9	14:47:22	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings

Object Info Session

MySQL Workbench

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Create a new schema in the connected server

SCHEMAS

Filter objects

- 2520090016_rahul
- bankdb
- joins_set_operations
- klu
- rahul_2520090016
- sys
- workflow_db

Limit to 1000 rows

```

91      'A fantastic underwater adventure aboard a mysterious submarine.',
92      1870
93  );
94  • SELECT * FROM Books;
95  • ALTER TABLE Books
96    ADD COLUMN embedding JSON;
97  • DESCRIBE Books;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI		auto_increment
title	varchar(200)	NO			
author	varchar(150)	NO			
genre	varchar(100)	YES			
description	text	YES			
published_year	int	YES			
embedding	json	YES			

Administration Schemas

Information

Schema:
2520090016_rahul

Result 4 x | Read On

Output

Action Output

#	Time	Action	Message
✓ 5	14:41:56	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
✓ 6	14:42:59	DESCRIBE Books	6 row(s) returned
✓ 7	14:44:26	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 8	14:44:57	SELECT * FROM Books LIMIT 0, 1000	10 row(s) returned
✓ 9	14:47:22	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 10	14:47:45	DESCRIBE Books	7 row(s) returned

Object Info Session

PART 5 — Add the mock vectors

MySQL Workbench interface showing a database connection named 'Rahul_2520090016'.

Navigator:

- SCHEMAS
 - Filter objects
 - 2520090016_rahul
 - bankdb
 - joins_set_operations
 - klu
 - rahul_2520090016
 - sys
 - workflow_db
- Administration
- Schemas
- Information

Schema: 2520090016_rahul

Query Editor: Practical week - 3*

```
98 • UPDATE Books
99 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
100 WHERE book_id = 1;
101
102 • UPDATE Books
103 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
104 WHERE book_id = 2;
105
106 • UPDATE Books
107 SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
108 WHERE book_id = 3;
109
110 • UPDATE Books
111 SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
112 WHERE book_id = 4;
113
114 • UPDATE Books
115 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
116 WHERE book_id = 5;
117
118 • UPDATE Books
119 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
120 WHERE book_id = 6;
```

Output:

Action Output

#	Time	Action	Message
✓ 15	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows mat
✓ 16	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows mat
✓ 17	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows mat
✓ 18	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows mat
✓ 19	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows mat
✓ 20	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows mat

Object Info Session

MySQL Workbench

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Practical week - 3*

SCHEMAS

Filter objects

2520090016_rahul

bankdb

joins_set_operations

klu

rahul_2520090016

sys

workflow_db

136 WHERE book_id = 10;

137 SELECT

138 book_id,

139 title,

140 embedding

141 FROM Books

142 ORDER BY book_id;

Result Grid

Filter Rows:

Edit: Export/Imports: Wrap Cell Contents:

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]
9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]

Administration Schemas

Information

Schema: 2520090016_rahul

Books 5 x

Output:

Action Output

#	Time	Action	Message
✓ 16	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 17	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 18	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 19	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 20	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 21	14:50:50	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned

Object Info Session

PART 6 — L2 Distance Search

MySQL Workbench

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- 2520090016_rahul
 - bankdb
 - joins_set_operations
 - klu
 - rahul_2520090016
 - sys
 - workflow_db

Practical week - 3*

Limit to 1000 rows

```

144 SELECT
145     book_id,
146     title,
147     author,
148     genre,
149
150     ROUND(
151         SQRT(
152             POW(
153                 CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) - 0.90,
154                 2
155             )
156         +
157         POW(
158             CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) - 0.80,

```

Administration Schemas

Information

Schema: 2520090016_rahul

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	Dune	Frank Herbert	Science Fiction	0.0866

Result 6 x

Output

Action Output

#	Time	Action	Message
17	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1
18	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1
19	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1
20	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1
21	14:50:50	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned
22	14:52:26	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(JSO...	3 row(s) returned

Object Info Session

PART 7 — Cosine Similarity

MySQL Workbench

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Practical week - 3

Limit to 1000 rows

SCHEMAS

Filter objects

- 2520090016_rahul
 - bankdb
 - joins_set_operations
 - klu
 - rahul_2520090016
 - sys
 - workflow_db

```
180 title,  
181 author,  
182 genre,  
183  
184 ROUND(  
185  
186 (  
187 CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) * 0.90  
188 +  
189 CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) * 0.80  
190 +  
191 CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) * 0.20  
192 )  
193
```

Administration Schemas

Information

Schema: 2520090016_rahul

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 7 x

Output

Action Output

#	Time	Action	Message
✓ 18	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 19	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 20	14:49:17	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 21	14:50:50	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned
✓ 22	14:52:26	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(USO...	3 row(s) returned
✓ 23	14:54:05	SELECT book_id, title, author, genre, ROUND((CAST(USON_EXTRACT(embeddin...	3 row(s) returned

Object Info Session